EMAN ALI KHAN **23P-0022**

PROJECT DETAILS

My Project Overview

In this project, I built a **real estate data analysis dashboard** for Peshawar housing listings from **Zameen.com**.

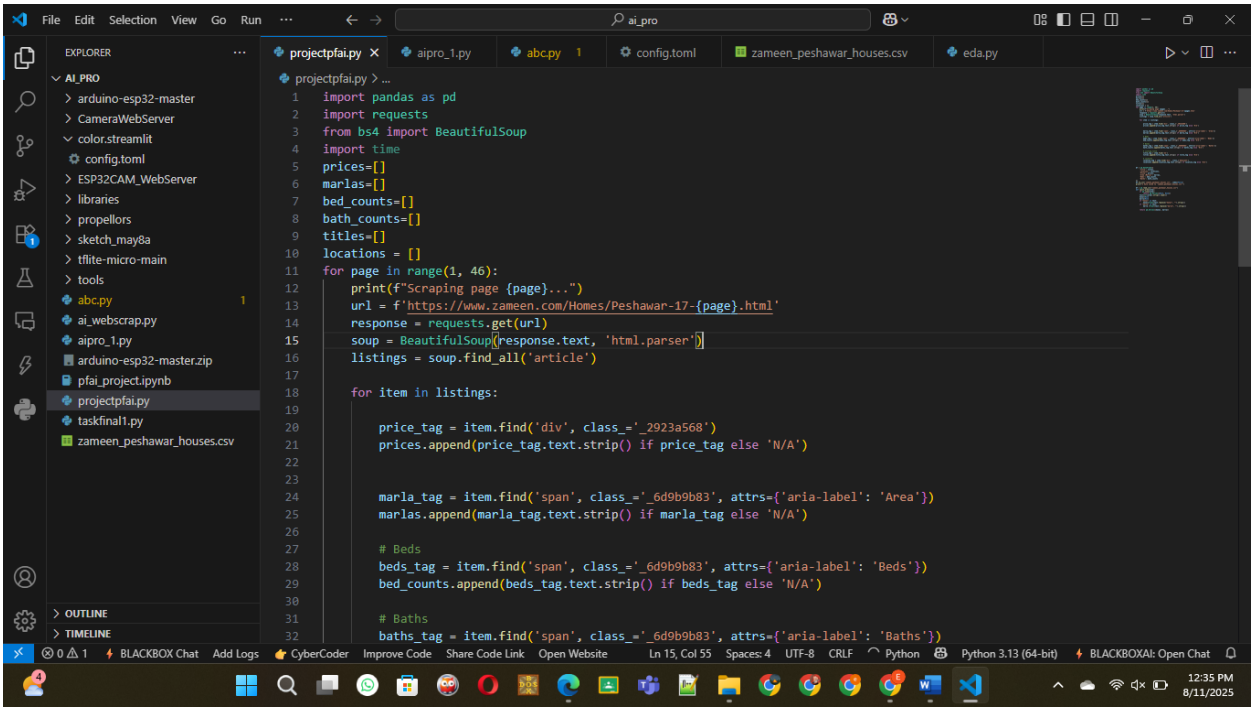
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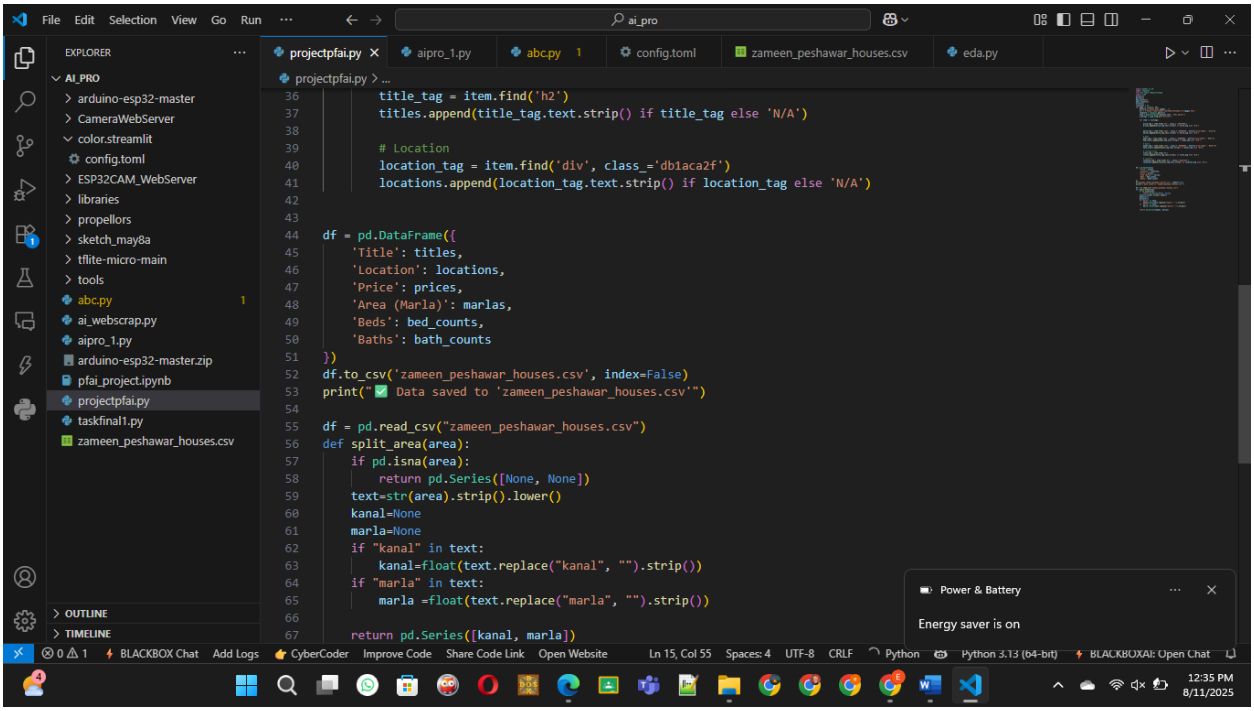
1. Web Scraping

First, I scraped house listings from Zameen.com. For each property, I collected details such as:

- Price
- Area (in Marla)
- Number of Beds
- Number of Baths
- Location

I stored the data in a CSV file (zameen_peshawar_houses.csv).



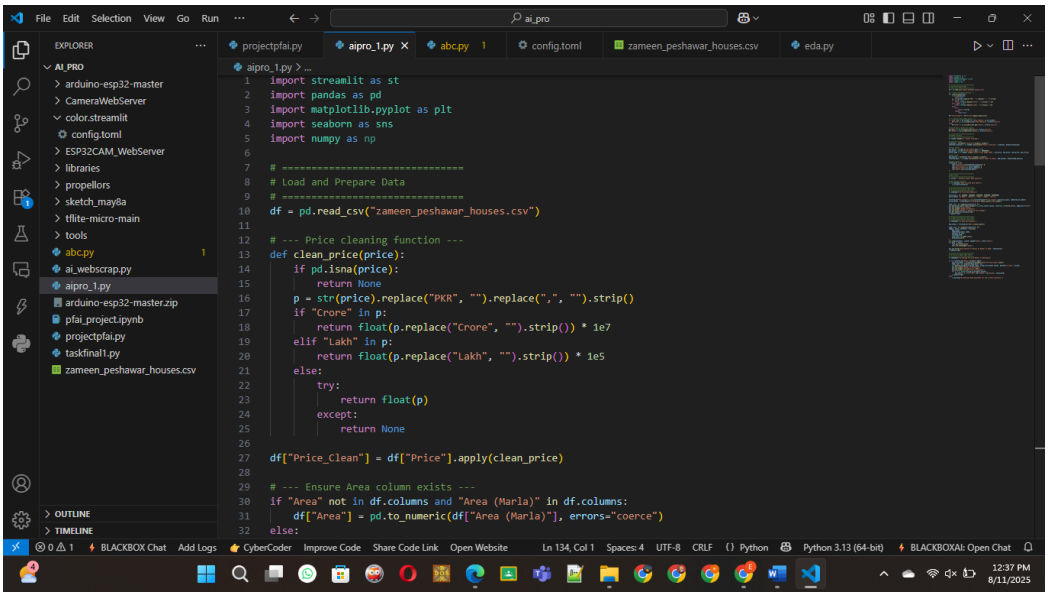


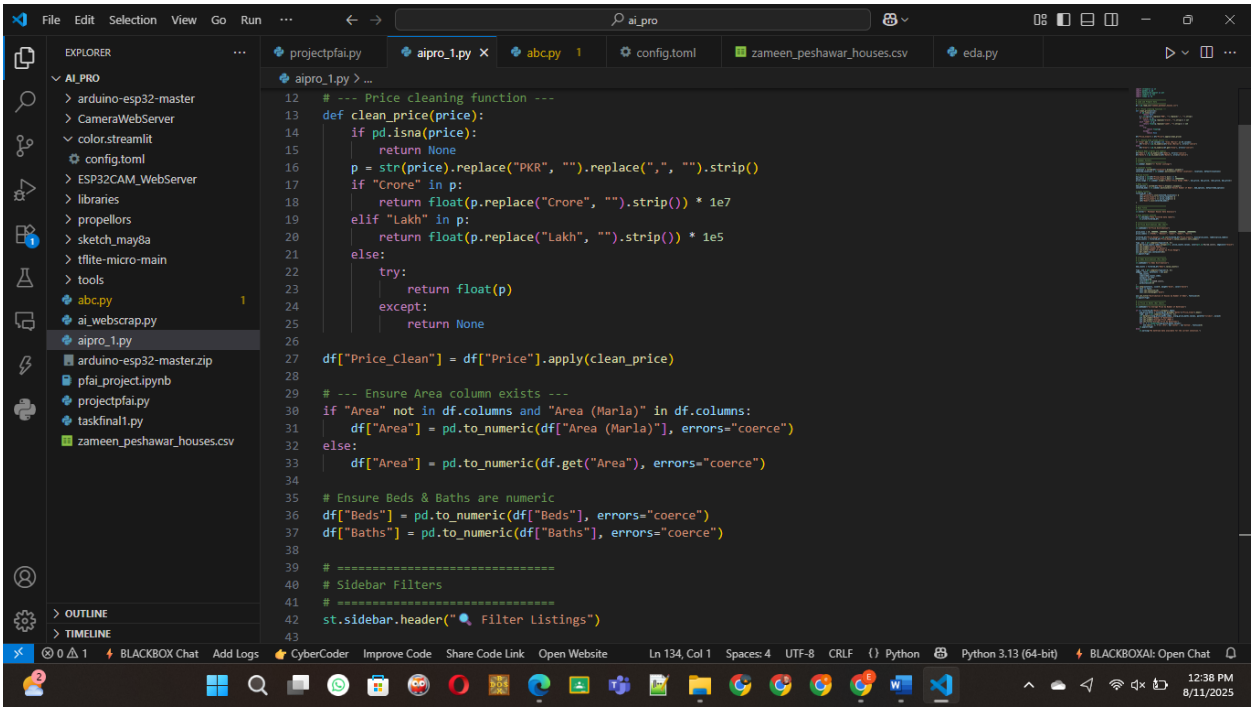
2. Data Cleaning & Preprocessing

The raw data had text, symbols, and inconsistent formats, so I cleaned it:

- Price: Removed “PKR”, commas, and converted Crore/Lakh values into actual numbers.
- Area: Made sure all areas were numeric (converted from strings like “5 Marla”).
- Beds & Baths: Converted them into numbers so I could analyze them.

This step ensured my dataset was ready for analysis.





```
# --- Price cleaning function ---
def clean_price(price):
    if pd.isna(price):
        return None
    p = str(price).replace("PKR", "").replace(",", "").strip()
    if "Crore" in p:
        return float(p.replace("Crore", "").strip()) * 1e7
    elif "Lakh" in p:
        return float(p.replace("Lakh", "").strip()) * 1e5
    else:
        try:
            return float(p)
        except:
            return None

df["Price_Clean"] = df["Price"].apply(clean_price)

# --- Ensure Area column exists ---
if "Area" not in df.columns and "Area (Marla)" in df.columns:
    df["Area"] = pd.to_numeric(df["Area (Marla)"], errors="coerce")
else:
    df["Area"] = pd.to_numeric(df.get("Area"), errors="coerce")

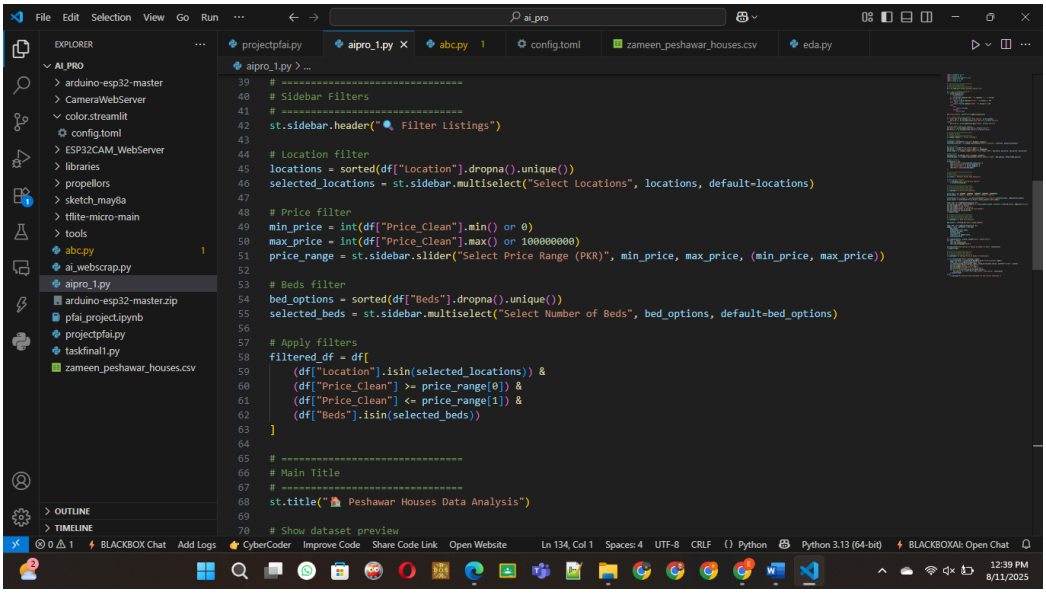
# Ensure Beds & Baths are numeric
df["Beds"] = pd.to_numeric(df["Beds"], errors="coerce")
df["Baths"] = pd.to_numeric(df["Baths"], errors="coerce")

# ================================
# Sidebar Filters
# ================================
st.sidebar.header("🔍 Filter Listings")
```

3. Streamlit Dashboard

I created an interactive Streamlit web app where users can:

- Filter houses by location, price range, and number of beds.
- Instantly see charts and stats update according to their selection.



```
# ================================
# Sidebar Filters
# ================================
st.sidebar.header("🔍 Filter Listings")

# Location filter
locations = sorted(df["Location"].dropna().unique())
selected_locations = st.sidebar.multiselect("Select Locations", locations, default=locations)

# Price filter
min_price = int(df["Price_Clean"].min() or 0)
max_price = int(df["Price_Clean"].max() or 100000000)
price_range = st.sidebar.slider("Select Price Range (PKR)", min_price, max_price, (min_price, max_price))

# Beds filter
bed_options = sorted(df["Beds"].dropna().unique())
selected_beds = st.sidebar.multiselect("Select Number of Beds", bed_options, default=bed_options)

# Apply filters
filtered_df = df[
    (df["Location"].isin(selected_locations)) &
    (df["Price_Clean"] >= price_range[0]) &
    (df["Price_Clean"] <= price_range[1]) &
    (df["Beds"].isin(selected_beds))
]

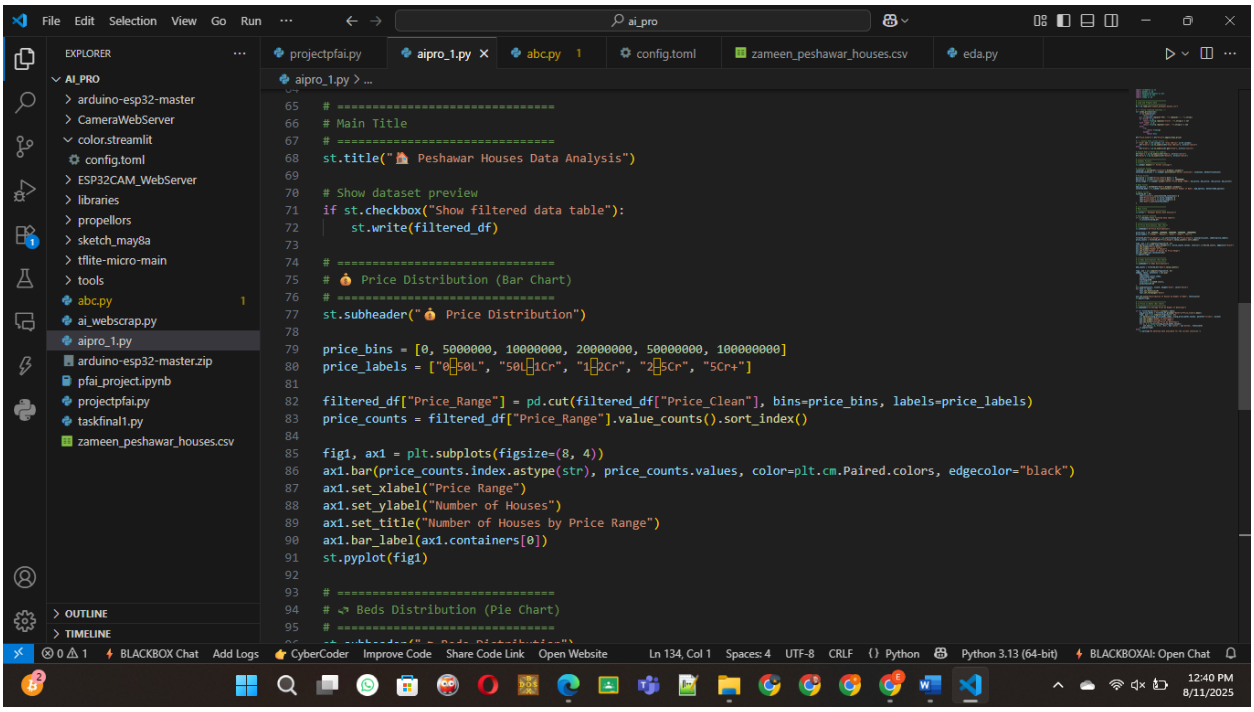
# ================================
# Main Title
# ================================
st.title("🏠 Peshawar Houses Data Analysis")

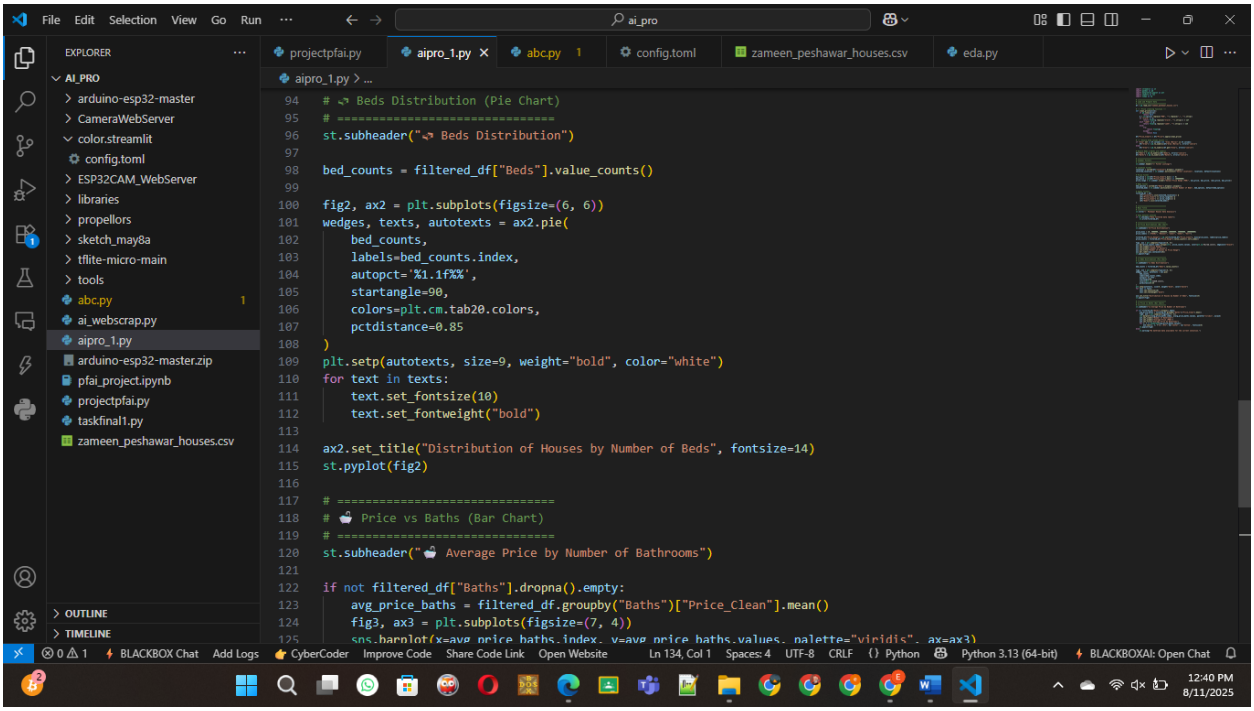
# Show dataset preview
```

4. Data Visualization

I used Plotly and Matplotlib charts in the dashboard:

- Price Distribution: Shows how many houses fall into different price ranges.
- Beds Distribution (Pie Chart): Shows the percentage of listings with different bedroom counts.
- Average Price by Bathrooms (Bar Chart): Displays the average price for listings with 1, 2, 3… bathrooms.





```
# 🛏️ Beds Distribution (Pie Chart)
# ==============================
st.subheader("🛏️ Beds Distribution")

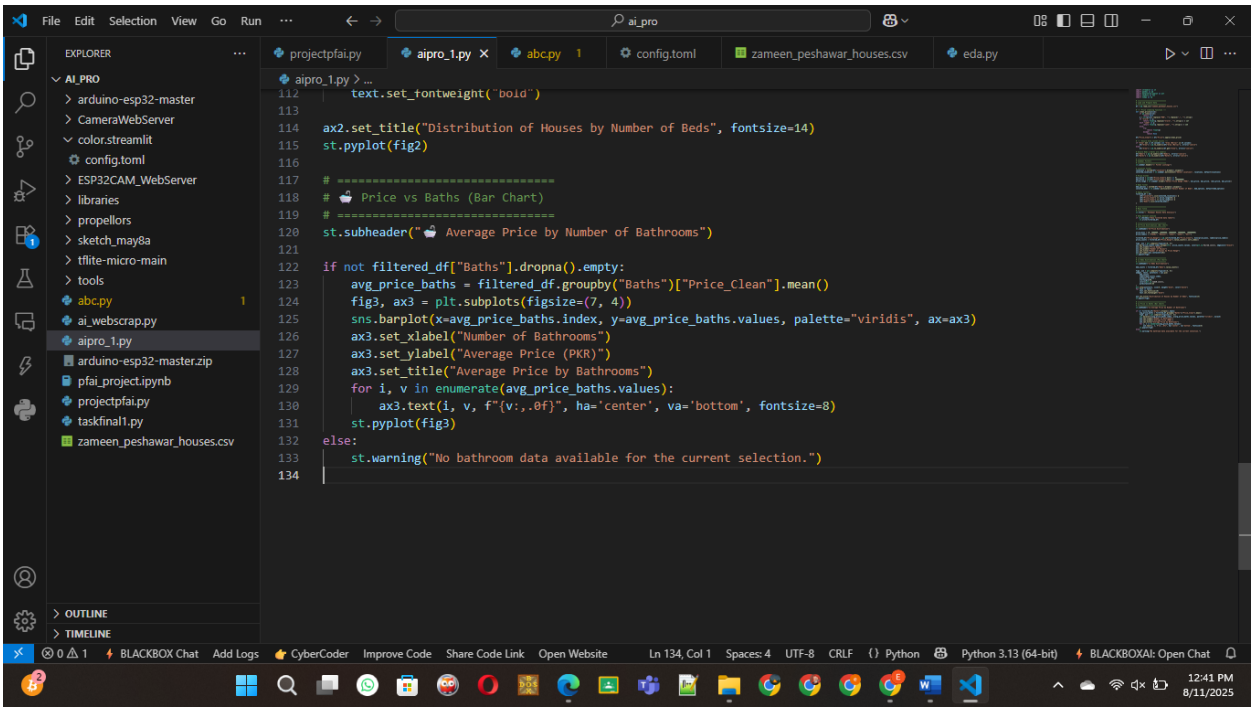
bed_counts = filtered_df["Beds"].value_counts()

fig2, ax2 = plt.subplots(figsize=(6, 6))
wedges, texts, autotexts = ax2.pie(
    bed_counts,
    labels=bed_counts.index,
    autopct='%1.1f%%',
    startangle=90,
    colors=plt.cm.tab20.colors,
    pctdistance=0.85
)
plt.setp(autotexts, size=9, weight="bold", color="white")
for text in texts:
    text.set_fontsize(10)
    text.set_fontweight("bold")

ax2.set_title("Distribution of Houses by Number of Beds", fontsize=14)
st.pyplot(fig2)

# ==============================
# 🛁 Price vs Baths (Bar Chart)
# ==============================
st.subheader("🛁 Average Price by Number of Bathrooms")

if not filtered_df["Baths"].dropna().empty:
    avg_price_baths = filtered_df.groupby("Baths")["Price_Clean"].mean()
    fig3, ax3 = plt.subplots(figsize=(7, 4))
    sns.barplot(x=avg_price_baths.index, y=avg_price_baths.values, palette="viridis", ax=ax3)
```



```
    text.set_fontweight("bold")

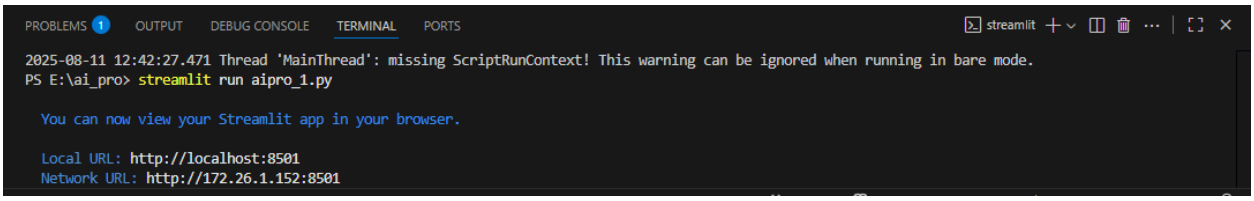
ax2.set_title("Distribution of Houses by Number of Beds", fontsize=14)
st.pyplot(fig2)

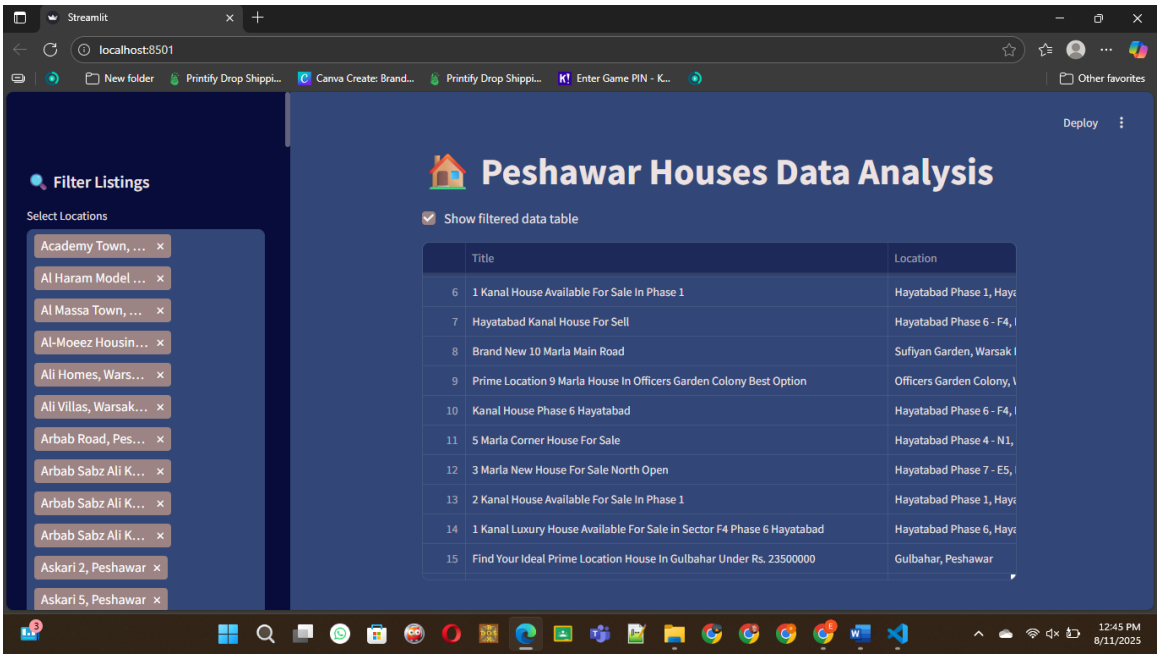
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st.subheader("🛁 Average Price by Number of Bathrooms")

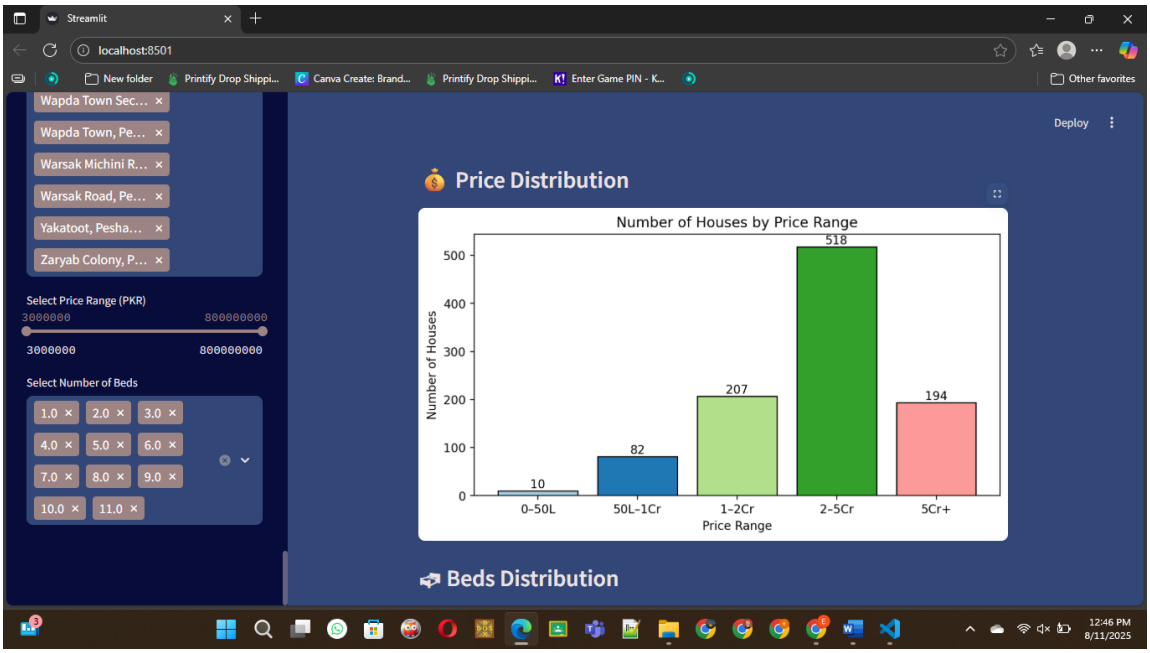
if not filtered_df["Baths"].dropna().empty:
    avg_price_baths = filtered_df.groupby("Baths")["Price_Clean"].mean()
    fig3, ax3 = plt.subplots(figsize=(7, 4))
    sns.barplot(x=avg_price_baths.index, y=avg_price_baths.values, palette="viridis", ax=ax3)
    ax3.set_xlabel("Number of Bathrooms")
    ax3.set_ylabel("Average Price (PKR)")
    ax3.set_title("Average Price by Bathrooms")
    for i, v in enumerate(avg_price_baths.values):
        ax3.text(i, v, f"{v:,.0f}", ha='center', va='bottom', fontsize=8)
    st.pyplot(fig3)
else:
    st.warning("No bathroom data available for the current selection.")
```

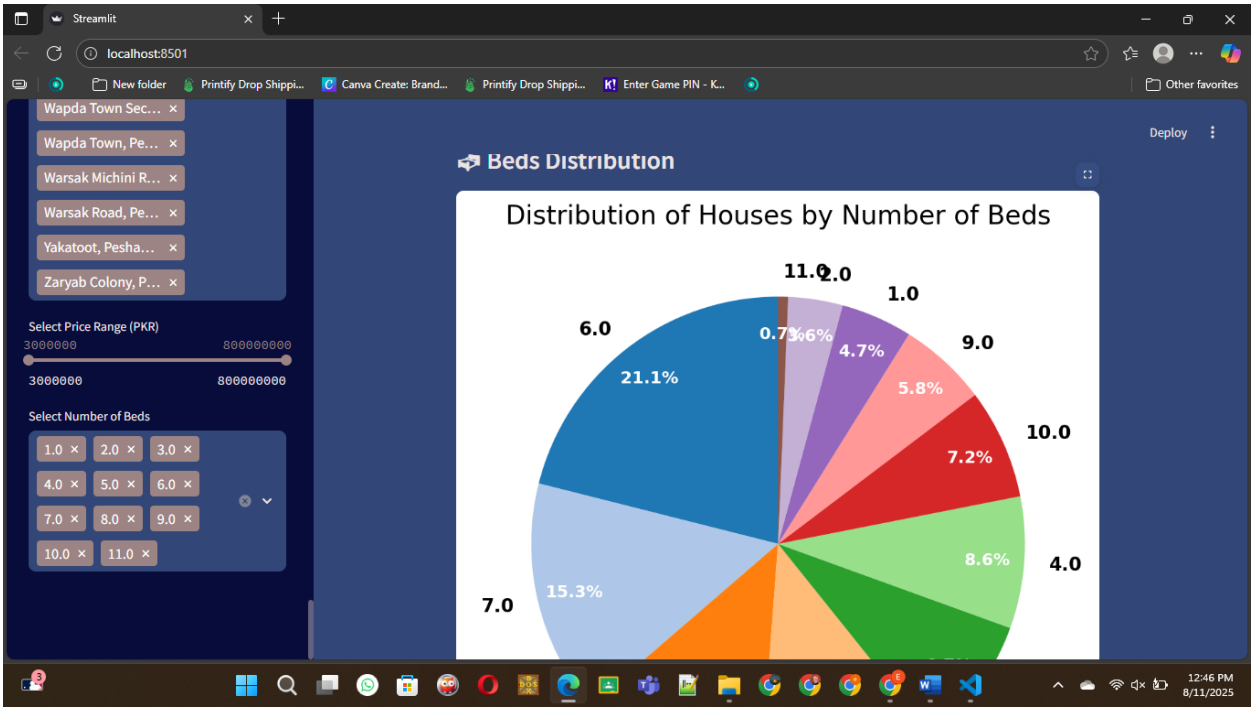
5. Final Outcome

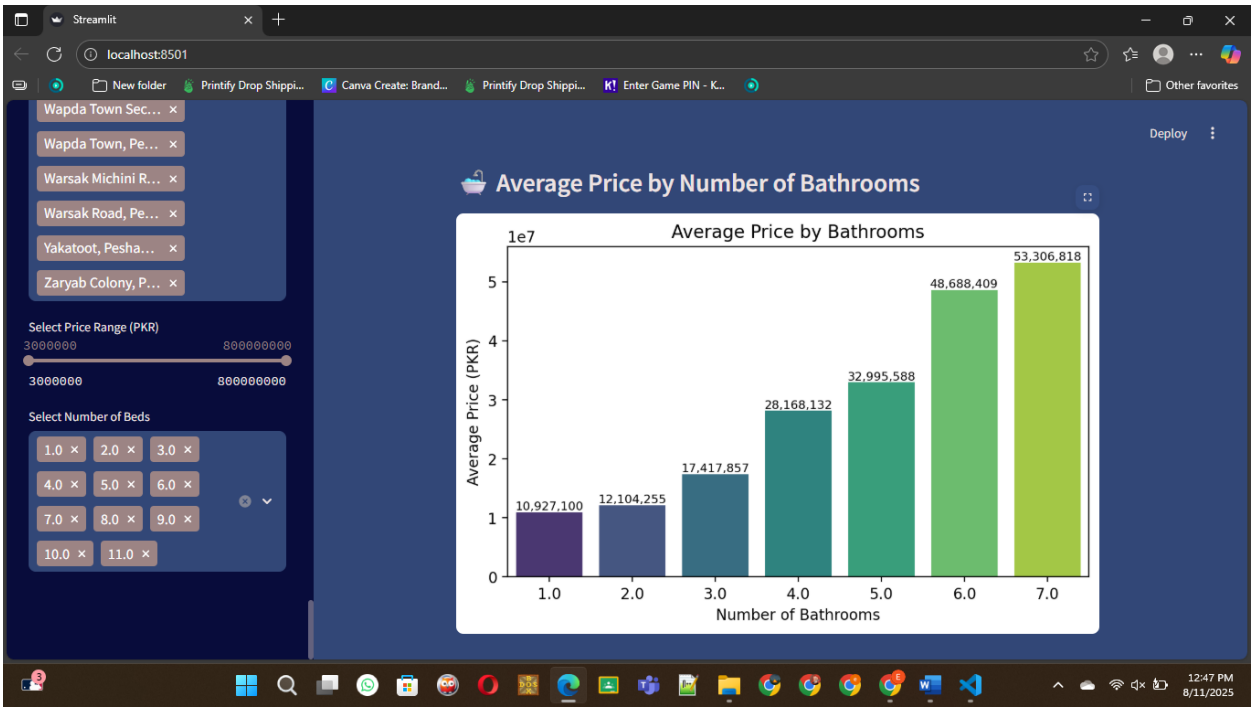
The final product is a visual, interactive dashboard that helps users explore the Peshawar housing market easily, without needing to read through every listing manually.








Streamlit
localhost:8501
Deploy
Wapda Town Sec...
Wapda Town, Pe...
Warsak Michini R...
Warsak Road, Pe...
Yakatoot, Pesha...
Zaryab Colony, P...
Select Price Range (PKR)
3000000
800000000
Select Number of Beds
1.0 2.0 3.0 4.0 5.0 6.0 7.0 8.0 9.0 10.0 11.0
Beds Distribution
Distribution of Houses by Number of Beds
11.0
2.0
1.0
6.0
9.0
10.0
4.0
7.0
0.7%
4.7%
5.8%
7.2%
8.6%
15.3%
21.1%
12:46 PM
8/11/2025


Streamlit
localhost:8501
Deploy
Wapda Town Sec...
Wapda Town, Pe...
Warsak Michini R...
Warsak Road, Pe...
Yakatoot, Pesha...
Zaryab Colony, P...
Select Price Range (PKR)
3000000
800000000
Select Number of Beds
1.0 2.0 3.0 4.0 5.0 6.0 7.0 8.0 9.0 10.0 11.0
Average Price by Number of Bathrooms
Average Price by Bathrooms
1e7
Average Price (PKR)
Number of Bathrooms
10,927,100
12,104,255
17,417,857
28,168,132
32,995,588
48,688,409
53,306,818
12:47 PM
8/11/2025